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A Horn clause is a clause with at most one non-negated literal. If all literals are negated, it is called a headless Horn clause. If one literal is non-negated, it is called a headed Horn clause.

Consider the following rule:

$\text{cunt}(x, y) : \text{female}(x), \text{sister}(x, z), \text{parent}(z, y)$ (Rule 1)
This means if x is a female and x is the sister of z and z is a parent of y , then x is the cunt of y .

Suppose we want to conclude sita is the cunt of mohan.

We have the following facts:

$\text{female}(\text{sita})$	$\text{--- } F_1$
$\text{sister}(\text{sita}, \text{geetha})$	$\text{--- } F_2$
$\text{parent}(\text{geetha}, \text{mohan})$	$\text{--- } F_3$

Instantiating the rule (1) we get

$\text{cunt}(\text{sita}, \text{mohan}) : \text{female}(\text{sita}), \text{sister}(\text{sita}, \text{geetha}), \text{parent}(\text{geetha}, \text{mohan})$.

Each F_1, F_2, F_3 is a headed Horn clause.

Rule (1) can be written in logical notation as:

$\forall x \forall y \forall z [(\text{female}(x) \wedge \text{sister}(x, z) \wedge \text{parent}(z, y)) \rightarrow \text{cunt}(x, y)]$

Using instantiation we get

$(\text{female}(\text{sita}) \wedge \text{sister}(\text{sita}, \text{geetha}) \wedge \text{parent}(\text{geetha}, \text{mohan})) \rightarrow \text{cunt}(\text{sita}, \text{mohan})$

Expressing as a clause this becomes

$\neg(\text{female}(\text{sita}) \wedge \text{sister}(\text{sita}, \text{geetha}) \wedge \text{parent}(\text{geetha}, \text{mohan})) \vee \text{cunt}(\text{sita}, \text{mohan})$